

CRISPIN JOE KENSLIN A

BIOINFORMATICS | BIOTECHNOLOGY

[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

Executive summary

- Biotechnology undergraduate specializing in **Bioinformatics, Structural Biology, and Computational Modelling**.
- Skilled in **Python, R, molecular docking**, MD simulations, and **machine learning**, integrating wet-lab and in silico approaches.
- Built multiple **bioinformatics tools** (clustering app, TF finder, hydrophobicity plotter) with **active GitHub deployment**.
- Conducted **in-silico SARS-CoV-2 inhibitor screening**, including docking and MD for prioritized phytochemicals.
- Experienced in **predictive modelling, sequence analysis**, and data **visualization** for biological datasets.
- Passionate about **AI-driven drug discovery** with a strong commitment to research and higher deeper learning.

Education

2022 - 2026

Tamil Nadu Agricultural University
B. Tech Biotechnology
(Bioinformatics module)
OGPA – 8.9/10.0

2009 - 2022

Sacred Heart Convent Anglo Indian Higher Secondary School
Class 1 to 12

Research interests

- Structural Bioinformatics
 - Machine Learning Models in Bioinformatics
 - Predictive Modelling & Data Analysis
 - Molecular Docking & Intermolecular Interactions
 - Sequence Analysis Tools and Algorithms
 - Computational Drug Discovery
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Technical Skills

Databases

- NCBI, KEGG, UniProt, PDB, Ensembl, PubChem, DrugBank, CTD, GEO, OMIM, BioCyc, and several crop-based databases.

Bioinformatics Tools

- BLAST, MEGA, RasMol, DNASP, MapMaker, ADMETlab, SwissADME, PyRx, Discovery Studio Visualizer, AutoDock, ChimeraX, SPDBV, OpenBabel, ClustalW, GROMACS.

Programming & Data Science

- **Python (Intermediate)** - Pandas, NumPy, Matplotlib, Scikit-learn

- **Machine Learning** – Linear/Logistic Regression, Polynomial Regression, Decision Trees, Random Forests, SVM, Markov Chains
- **R (Advanced)** - Statistical Analysis, Hypothesis Testing, PCA, Correlation, Regression, Data Visualization (ggplot2, Tidyverse, dplyr, Shiny)
- **Other Languages:**
C (Intermediate), Perl (Intermediate)

Operating Systems

- **Linux (Advanced):** Command-line operations, shell scripting, environment setup, package management, and server-based data processing.
- **Windows** (Proficient) and macOS.

Version Control Systems

- Git and GitHub

Soft Skills

- Leadership
- Project Management
- Team Work
- Time Management
- Quick Learning

Projects

- **In-silico Analysis** of Inhibitory Phytochemicals of *Oenothera biennis* against Multiple Targets of **SARS-CoV-2** (Ongoing)
 - Performed Molecular docking of around 70 phytochemicals of evening primrose against 5 targets of SARS-COV-2 and performed molecular dynamics for the best compounds with the complexes.
- **Clustrix Pro** - "An interactive web application to explore and visualize data patterns with precise clustering"
 - Developed a web application using R and Shiny with an easy-to-use interface that performs data clustering with multiple algorithms and methods, provides publication-ready plots and diagrams and a detailed report on the analysis with clear interpretations and further prospects.
- **Transcription Factor Finder Pro**
 - A web-based Transcription finder application that accepts DNA sequences and finds the sites of transcription factors and output into a visual diagram and table format which can be downloaded. ([Link](#))
- **Tumor Prediction Model**
 - Developed a beginner-friendly machine learning model, Tumor Model, using Scikit-learn's RandomForestClassifier to predict tumor presence from gene expression data sourced from Kaggle's Genomic Data for Cancer. ([Link](#))
- **DNA Sequence Analyzer**
 - Developed a Python-based sequence analysis tool from scratch.
- **Hydrophobicity Plotter** – Multi-window graphical hydrophobicity analysis tool. ([Link](#))
- **RNA Expression Analysis Script** – Simulated RNA expression data analysis in R. ([Link](#))

Research Engagements & Conferences

- **Ecofest'24** (International Conference): Best Poster Award – **Predictive Analysis using AI in Energy and Environment**.
- **Bioconcorrenza'23 & '24**: National Symposium on Biotechnology.
- **Mendel's Day 2025**: Presented poster on The **Legacy of Gregor Mendel**.

Academic Engagements & Workshops

- Interactive session with **Dr. Sudhagar Pitchaimuthu** (Heriot-Watt University, Switzerland) – “**What after Biotech UG?**”
- **Workshop**: Agri-Tech Startups Business Meet 2025, COXBIT, TNAU
- **Guest Lectures** attended:
 - **Dr. Azhaguvel Perumal** – Trait Development & Commercialization in seed industry
 - **Dr. Vivek Chandramohan** – Career Opportunities in Omics Data
 - **Dr. Ajay Kohli** – IRRI, India
 - **Dr. Jagadeesh Selvam** – Molecular Breeding & Genotyping in the seed industry
 - **Dr. Sumanth Mutte** – AI & Big Data in Plant Genomics
 - **Dr. Preethi Marimuthu** – Biotech Solutions for Climate Change. Carbon capture techniques

Certifications

2025	Machine Learning I – Columbia University (Columbia+ Online)
2024	Nanotechnology in Agriculture – NPTEL, IIT Kanpur (8 weeks) – Elite Certificate
2024	Experimental Biotechnology – NPTEL, IIT Guwahati (12 weeks) – Elite Certificate

Awards

2024	Best Poster Presentation Award – Ecofest'24
2022	Subject Topper in Biology – Class 12

Languages

Tamil	Native
English	Proficient (School and college medium)
French	Intermediate

Extra-curriculars

- Music: Keyboard, Guitar, Drums (10+ years)
- Drawing and Art
- Photography & Filmography: Editing, Animation, 3D Modelling
- Electronics & Physics Enthusiast
- Sports: Basketball, Table Tennis
- NSS Volunteer